

Network Time Protocol

- Topics covered:
 - Why Is Time Important for Network Devices?
 - Manual Time Configuration
 - Network Time Protocol Basics
 - NTP Configuration

Why Is Time Important for Network Devices?

Importance of Time on Network Devices

All network devices (routers, switches, PCs, smartphones) maintain an **internal clock**. This clock is used to track system time and is essential for operations like logging and event tracking.

In Cisco IOS, the time can be checked using:

- `show clock` → displays current time
- `show clock detail` → displays time + source

```
R1#show clock
*00:16:00.857 UTC Sat Dec 26 2020
```

```
R1#show clock detail
*00:19:49.411 UTC Sat Dec 26 2020
Time source is hardware calendar
```

Explanation:

- The default timezone is **UTC (Coordinated Universal Time)**
- The device uses the **hardware calendar (internal clock)** as the default time source
- The * symbol indicates the time is not authoritative
- This means the device does not fully trust the accuracy of its time

The reason is that internal clocks drift over time, so relying only on them can lead to inaccurate time.

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Why Accurate Time Matters (Logs & Troubleshooting)

Network devices store logs of important events such as:

- Interface status changes
- Routing neighbor relationships (e.g., OSPF)
- Device restarts

These logs include timestamps, so correct time is critical.

CLI Output (R2 Logs)

```
R2#show logging
!output abbreviated!
*Dec 27 00:50:20.005: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.122.192 on GigabitEthernet0/0 from LOADING to FULL, Loading Done
*Dec 27 01:06:38.653: %OSPF-5-ADJCHG: Process 1, Nbr 10.0.0.6 on GigabitEthernet0/1 from LOADING to FULL, Loading Done
*Dec 27 01:07:07.311: %OSPF-5-ADJCHG: Process 1, Nbr 10.0.0.6 on GigabitEthernet0/1 from LOADING to FULL, Loading Done
*Dec 27 01:08:29.924: %OSPF-5-ADJCHG: Process 1, Nbr 10.0.0.6 on GigabitEthernet0/1 from FULL to DOWN, Neighbor Down: Dead timer expired
*Dec 27 01:09:10.714: %OSPF-5-ADJCHG: Process 1, Nbr 10.0.0.6 on GigabitEthernet0/1 from LOADING to FULL, Loading Done

R2#show clock
*01:17:06.706 UTC Sun Dec 27 2020
```

CLI Output (R3 Logs)

```
R3#show logging
!output abbreviated!
May 23 16:24:17.320: %OSPF-5-ADJCHG: Process 1, Nbr 10.0.0.5 on GigabitEthernet0/0 from LOADING to FULL, Loading Done
May 23 16:25:08.758: %OSPF-5-ADJCHG: Process 1, Nbr 10.0.0.5 on GigabitEthernet0/0 from FULL to DOWN, Neighbor Down: Interface down or detached
May 23 16:25:10.714: %LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to down
May 23 16:25:11.716: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to down
May 23 16:26:14.976: %LINK-3-UPDOWN: Interface GigabitEthernet0/0, changed state to up
May 23 16:26:15.977: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
May 23 16:26:20.618: %OSPF-5-ADJCHG: Process 1, Nbr 10.0.0.5 on GigabitEthernet0/0 from LOADING to FULL, Loading Done

R3#show clock
16:30:37.020 UTC Fri May 23 2008
```

Explanation

When troubleshooting, logs from multiple devices must be compared.

- R2 shows events around recent time
- R3 shows events in a completely different year

This mismatch makes it difficult to correlate events across devices.

Even though the issue (interface going down) can still be identified here, in real networks with many logs, incorrect time makes troubleshooting much more complex and time-consuming.

Manual Time Configuration

Configuring Software Clock

The software clock can be manually configured using the clock set command.

CLI Output

```
R2#clock set ?
hh:mm:ss Current Time
R2#clock set 14:30:00 ?
<1-31> Day of the month
MONTH Month of the year
R2#clock set 14:30:00 27 ?
MONTH Month of the year
R2#clock set 14:30:00 27 Dec ?
<1993-2035> Year
R2#clock set 14:30:00 27 Dec 2020 ?
<cr>
R2#clock set 14:30:00 27 Dec 2020

R2#show clock detail
14:30:05.887 UTC Sun Dec 27 2020
Time source is user configuration
```

Explanation

The command is entered step-by-step:

1. Enter time → hh:mm:ss
2. Enter day or month (either order allowed)
3. Enter remaining month/day
4. Enter year
5. Press Enter (<cr>)

After configuration:

- Time source becomes user configuration
- This is done in privileged EXEC mode
- It is not saved in running configuration

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Hardware Clock (Calendar) Configuration

The device also has a hardware clock called the **calendar**, which is separate from the software clock.

CLI Output

```
R2#calendar set 14:35:00 ?
<1-31> Day of the month
MONTH Month of the year

R2#calendar set 14:35:00 27 ?
MONTH Month of the year

R2#calendar set 14:35:00 27 Dec ?
<1993-2035> Year

R2#calendar set 14:35:00 27 Dec 2020 ?
<cr>

R2#calendar set 14:35:00 27 Dec 2020
```

```
R2#show calendar
14:35:07 UTC Sun Dec 27 2020
```

Explanation

- calendar refers to the hardware clock
- clock refers to the software clock

Both are independent, meaning:

- You can configure each separately
- They may have different times if not synchronized

Synchronizing Clock and Calendar

Since both clocks are separate, they should normally be synchronized.

Case 1: Sync Calendar to Clock

```
R2#show clock
14:38:14.301 UTC Sun Dec 27 2020
R2#show calendar
00:00:03 UTC Sun Dec 27 2020
R2#clock update-calendar
R2#show clock
14:38:22.181 UTC Sun Dec 27 2020
R2#show calendar
14:38:23 UTC Sun Dec 27 2020
```


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Case 2: Sync Clock to Calendar

```
R2#show clock
00:00:15.788 UTC Mon Sep 6 1993
R2#show calendar
14:55:07 UTC Sun Dec 27 2020
R2#clock read-calendar
R2#show clock
14:55:12.522 UTC Sun Dec 27 2020
R2#show calendar
14:55:15 UTC Sun Dec 27 2020
```

Explanation:

- clock update-calendar → copies **clock** → **calendar**
- clock read-calendar → copies **calendar** → **clock**

Keeping both synchronized ensures consistency before applying further configurations.

Configuring Time Zone

Timezone defines how the device displays time relative to UTC.

CLI Output

```
R2(config)#do show clock
15:13:33.985 UTC Sun Dec 27 2020
R2(config)#clock timezone ?
WORD name of time zone

R2(config)#clock timezone JST ?
<-23 - 23> Hours offset from UTC

R2(config)#clock timezone JST 9 ?
<0-59> Minutes offset from UTC
<cr>

R2(config)#clock timezone JST 9
R2(config)#do show clock
00:13:45.414 JST Mon Dec 28 2020
R2(config)#do clock set 15:15:00 Dec 27 2020
R2(config)#do show clock
15:15:02.129 JST Sun Dec 27 2020
```

Explanation

- Command format:
- clock timezone name hours-offset [minutes-offset]
- Configured in global configuration mode
- Stored in the running configuration

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When timezone is changed:

- Time label changes (UTC → JST)
- Time value automatically adjusts (e.g., +9 hours)

Since actual local time may differ after adjustment, the clock may need to be set again manually. Also, NTP operates in UTC, so timezone configuration is necessary for correct local display.

Daylight Saving Time (Summer Time)

Some countries adjust time forward and backward during the year.

Concept (Canada Example)

Start of DST: Second Sunday of March at 02:00

End of DST: First Sunday of November at 02:00

 Canada	Northern America	Northern	Second Sunday March at 02:00 local standard time (for most of Canada)	First Sunday November at 02:00 local daylight saving time (for most of Canada)
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CLI Configuration (Step-by-Step)

```
R2(config)#clock summer-time ?  
WORD name of time zone in summer  
R2(config)#clock summer-time EDT ?  
date Configure absolute summer time  
recurring Configure recurring summer time  
R2(config)#clock summer-time EDT recurring ?  
<1-4> Week number to start  
first First week of the month  
last Last week of the month  
<cr>
```

```
R2(config)#clock summer-time EDT recurring 2 ?  
DAY Weekday to start  
R2(config)#clock summer-time EDT recurring 2 Sunday ?  
MONTH Month to start  
R2(config)#clock summer-time EDT recurring 2 Sunday March ?  
hh:mm Time to start (hh:mm)  
R2(config)#clock summer-time EDT recurring 2 Sunday March 02:00 ?  
<1-4> Week number to end  
first First week of the month  
last Last week of the month
```

```
R2(config)#clock summer-time EDT recurring 2 Sunday March 02:00 1 ?  
DAY Weekday to end  
R2(config)#clock summer-time EDT recurring 2 Sunday March 02:00 1 Sunday ?  
MONTH Month to end  
R2(config)#clock summer-time EDT recurring 2 Sunday March 02:00 1 Sunday November ?  
hh:mm Time to end (hh:mm)  
R2(config)#clock summer-time EDT recurring 2 Sunday March 02:00 1 Sunday November 02:00 ?  
<1-1440> Offset to add in minutes  
<cr>  
R2(config)#clock summer-time EDT recurring 2 Sunday March 02:00 1 Sunday November 02:00
```

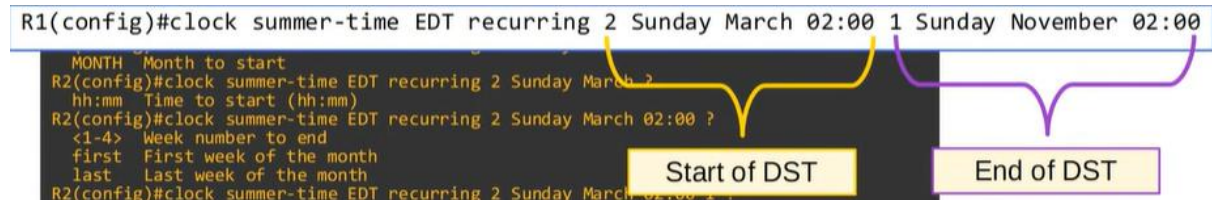
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Breakdown

Start: 2 Sunday March 02:00

End: 1 Sunday November 02:00

```
R1(config)#clock summer-time EDT recurring 2 Sunday March 02:00 1 Sunday November 02:00
MONTH Month to start
R2(config)#clock summer-time EDT recurring 2 Sunday March 02:00
hh:mm Time to start (hh:mm)
R2(config)#clock summer-time EDT recurring 2 Sunday March 02:00 ?
<1-4> Week number to end
first First week of the month
last Last week of the month
R2(config)#clock summer-time EDT recurring 2 Sunday March 02:00 1 Sunday November 02:00
```



Explanation

- Configured in global configuration mode
- recurring ensures automatic yearly adjustment
- Order must be followed strictly:
 - week → weekday → month → time

Additional detail:

- Default offset is 60 minutes
- Long commands may wrap in CLI using \$, but represent one command

Manual Time Configuration

```
R1# show clock
```

```
R1# show clock detail
```

```
R1# clock set hh:mm:ss {day|month} {month|day} year
```

```
R1# show calendar
```

```
R1# calendar set hh:mm:ss {day|month} {month|day} year
```

```
R1(config)# clock timezone name hours-offset [minutes-offset]
```

```
R1(config)# clock summer-time recurring name start end [offset]
```

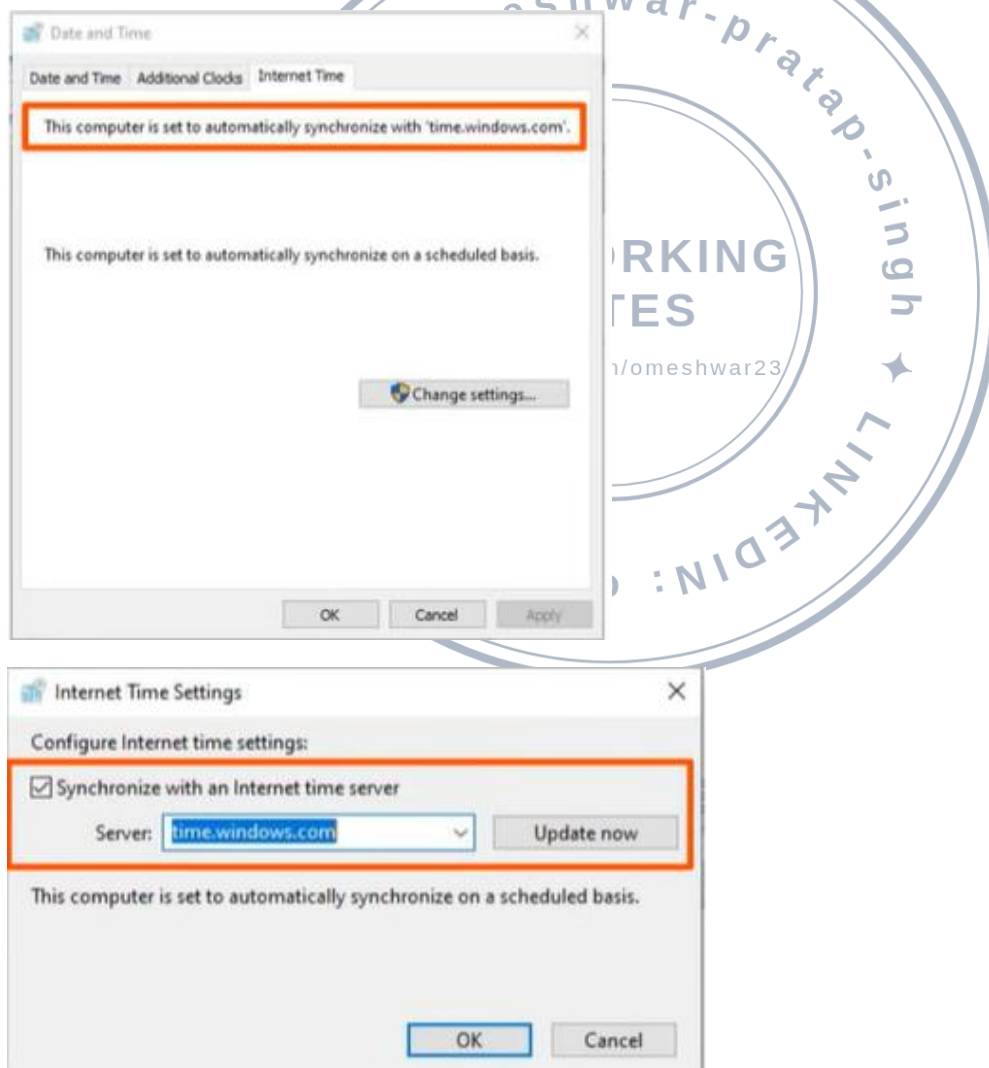
Network Time Protocol Basics

Introduction to NTP (Network Time Protocol)

Previously, time was configured manually, but this approach does not scale in real networks. Configuring time individually on every router, switch, firewall, PC, and phone would take significant effort, and even then, the clocks would gradually drift and become inaccurate.

To solve this, **NTP (Network Time Protocol)** is used. It allows devices to automatically synchronize their time with reliable NTP servers over a network.

A common real-world example is a PC automatically syncing its time.



This shows that instead of relying on its internal clock, the system regularly synchronizes with an external **NTP server on the Internet**.

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Resolving NTP Server Names using DNS (Beyond Topic)

Before a device can contact an NTP server, it must first resolve the server's domain name into an IP address using DNS.

CLI Output:

```
C:\Users\user>nslookup time.windows.com
Server:  dns.google
Address:  8.8.8.8

Non-authoritative answer:
Name:     time.microsoft.akadns.net
Address:  20.43.94.199
Aliases:  time.windows.com

C:\Users\user>nslookup time.google.com
Server:  dns.google
Address:  8.8.8.8

Non-authoritative answer:
Name:     time.google.com
Addresses: 2001:4860:4806::
           2001:4860:4806:c::
           2001:4860:4806:8::
           2001:4860:4806:4::
           216.239.35.12
           216.239.35.8
           216.239.35.4
           216.239.35.0
```

Explanation

The device sends a query to a DNS server (in this case 8.8.8.8) to resolve the hostname.

- time.windows.com resolves to a single IPv4 address
- time.google.com resolves to multiple IPv4 and IPv6 addresses

This step is necessary so the device knows where to send NTP requests.

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How NTP Operates

Once the IP address is known, the device can communicate with NTP servers.

- Devices request time from NTP servers and synchronize automatically
- A device can act as:
 - Client (request time)
 - Server (provide time)
 - Or both at the same time

Accuracy depends on distance:

- Same LAN → about 1 ms
- Over Internet/WAN → about 50 ms

NTP uses UDP port 123 for communication.

Reference Clocks and Stratum Concept

At the core of NTP are reference clocks, which are highly accurate time sources such as atomic or GPS clocks.

Diagram:



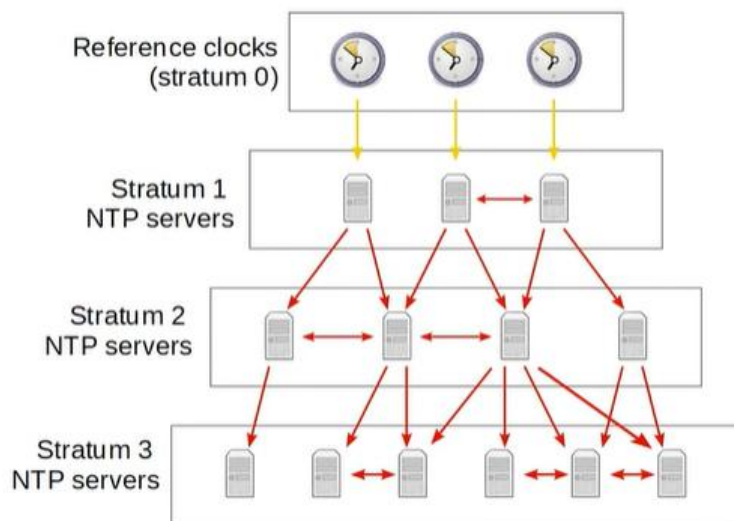
These are considered **Stratum 0**, meaning they are the original time sources.

Devices cannot directly use these clocks, but they can use servers connected to them.

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NTP Hierarchy (Stratum Levels)

Time is distributed through a hierarchy based on stratum levels.



Explanation:

- Stratum 1 servers get time directly from reference clocks
- Stratum 2 and 3 servers get time from other NTP servers
- As stratum increases, accuracy slightly decreases

Maximum usable stratum is **15**. Anything higher is considered unreliable and will not be used.

Primary and Secondary Servers

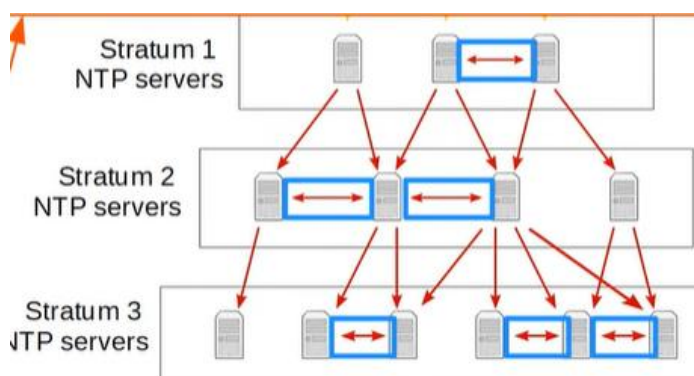
The hierarchy also defines server roles.

Primary Servers (Stratum 1) ← Direct connection to reference clocks

Secondary Servers (Stratum 2+) ← Get time from other NTP servers

NTP Peering and Modes

Devices can also exchange time with other devices at the same level.



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This is called **peering (symmetric active mode)** and provides:

- Better accuracy
- Redundancy if a lower-stratum server fails

NTP supports three modes:

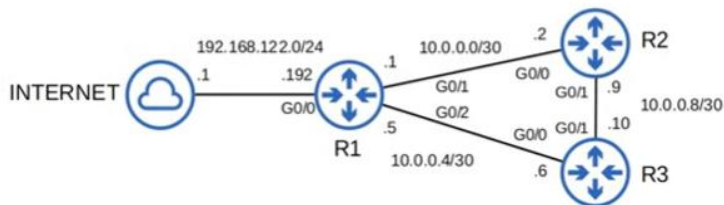
- Server mode
- Client mode
- Symmetric active mode

A device can operate in all modes at the same time.



NTP Configuration

Network Topology for Configuration



In this setup:

- R1 connects to the Internet
- R1 synchronizes with external NTP servers
- R2 and R3 can later synchronize from R1

Configuring NTP Servers

```
C:\Users\user>nslookup time.google.com
Server:  dns.google
Address:  8.8.8.8

Non-authoritative answer:
Name:     time.google.com
Addresses: 2001:4860:4806::
           2001:4860:4806:c::
           2001:4860:4806:8::
           2001:4860:4806:4::
           216.239.35.12
           216.239.35.8
           216.239.35.4
           216.239.35.0
```

After obtaining server IPs, they are configured on the router.

CLI Output

```
R1(config)#ntp server 216.239.35.0
R1(config)#ntp server 216.239.35.4
R1(config)#ntp server 216.239.35.8
R1(config)#ntp server 216.239.35.12
```

Explanation

Multiple servers are configured to ensure:

- Reliability
- Better response selection

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The router queries all servers and dynamically chooses the best one.

The selected server may change over time depending on response quality.

Now, if you want to manually make the device prefer one of the configured servers, you can add PREFER to the end of the command

```
R1(config)#ntp server 216.239.35.0 prefer
R1(config)#ntp server 216.239.35.4
R1(config)#ntp server 216.239.35.8
R1(config)#ntp server 216.239.35.12
```

And others will be backups, But for this we not using PREFER keyword.

Verifying NTP Associations

```
R1#show ntp associations

address      ref clock      st  when  poll reach  delay  offset  disp
*~216.239.35.0 .GOOG.         1   43    64   17 62.007 1401.54 0.918
+~216.239.35.8 .GOOG.         1   43    64   17 64.220 1416.65 0.939
+~216.239.35.4 .GOOG.         1   47    64   17 57.669 1402.11 0.916
+~216.239.35.12 .GOOG.         1   39    64   17 62.229 1409.03 0.960
* sys.peer, # selected, + candidate, - outlyer, x falseticker, ~ configured
```

Explanation

Each symbol indicates status:

- → current synchronization server (sys.peer)
- + → candidate servers
- ~ → configured servers

All servers shown are stratum 1, meaning they are close to the reference source.

The selected server may change dynamically as the router evaluates responses.

Checking NTP Status

```
R1#show ntp status
Clock is synchronized, stratum 2, reference is 216.239.35.12
nominal freq is 1000.0003 Hz, actual freq is 999.5003 Hz, precision is 2**14
ntp uptime is 295800 (1/100 of seconds), resolution is 1001
reference time is E393F0A9.1F758C5B (05:50:33.122 UTC Mon Dec 28 2020)
clock offset is 1343.7280 msec, root delay is 49.13 msec
root dispersion is 2275.31 msec, peer dispersion is 3.44 msec
loopfilter state is 'SPIK' (Spike), drift is 0.000499999 s/s
system poll interval is 64, last update was 173 sec ago.
```

Explanation

- The clock is successfully synchronized
- Router becomes **stratum 2** because it syncs from stratum 1 servers
- The reference IP shows which server is currently used

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Timezone and Calendar with NTP

Let's Check R1 clock and calendar again

```
R1(config)#do show clock detail
06:56:32.315 UTC Mon Dec 28 2020
Time source is NTP
```

```
R1(config)#do show calendar
05:23:06 UTC Mon Dec 28 2020
```

```
R1(config)#clock timezone JST 9
R1(config)#ntp update-calendar
R1(config)#do show clock detail
15:57:33.078 JST Mon Dec 28 2020
Time source is NTP
R1(config)#do show calendar
15:57:36 JST Mon Dec 28 2020
```

Explanation:

After NTP synchronization:

- The **software clock** is correct but still in UTC
- The **calendar (hardware clock)** is not updated automatically

To fix this:

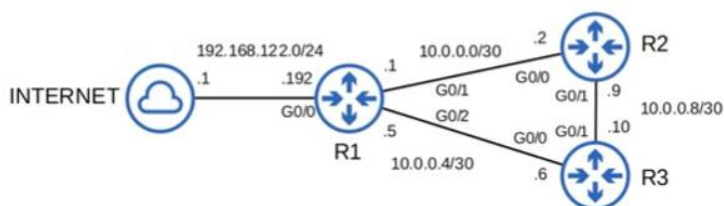
- Configure timezone (e.g., JST)
- Use `ntp update-calendar` to sync hardware clock

The hardware clock is important because it:

- Maintains time during reboot or power loss
- Initializes the software clock at startup

Using a Router as an NTP Server (R1 → R2)

In smaller networks, devices usually sync directly with public NTP servers. However, here R2 is configured to use R1 as its NTP server.



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Before doing that, a loopback interface is configured on R1.

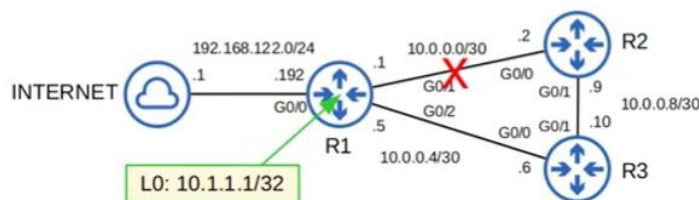
```
R1(config)#interface loopback0
R1(config-if)#ip address 10.1.1.1 255.255.255.255
R1(config-if)#exit
R1(config)#ntp source loopback0
```

Explanation

- A loopback interface provides a stable IP address
- ntp source loopback0 ensures NTP packets use 10.1.1.1 as source

Why this is important:

- If a physical interface (e.g., G0/1) fails → NTP breaks
- Loopback remains reachable via routing → NTP continues working



Configuring R2 to Use R1 as NTP Server

After preparing R1, R2 is configured to sync with it.

```
R2(config)#ntp server 10.1.1.1
R2(config)#do show ntp associations

address      ref clock      st  when  poll reach  delay  offset  disp
*~10.1.1.1    216.239.35.12  2   0     64    1  7.038 -13.128 3937.5
* sys.peer, # selected, + candidate, - outlyer, x falseticker, ~ configured
R2(config)#do show ntp status
Clock is synchronized, stratum 3, reference is 10.1.1.1
...
```

Explanation

- indicates R2 is **synchronizing with R1**
- R1's reference is a Google NTP server (stratum 1)

Therefore:

- R1 becomes **stratum 2**
- R2 becomes **stratum 3**

This shows how stratum increases as time moves further from the original source.

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Configuring Multiple NTP Servers on R3

```
R3(config)#ntp server 10.1.1.1
R3(config)#ntp server 10.2.2.2
R3(config)#do show ntp associations
```

address	ref clock	st	when	poll	reach	delay	offset	disp
*~10.1.1.1	216.239.35.0	2	1	64	0	0.000	0.000	15937.
~10.2.2.2	10.1.1.1	3	1	64	0	0.000	0.000	15937.

* sys.peer, # selected, + candidate, - outlyer, x falseticker, ~ configured

Explanation:

- R3 prefers R1 (10.1.1.1) over R2
- Reason:
 - R1 has lower stratum (2)
 - R2 has higher stratum (3)

This confirms an important rule:

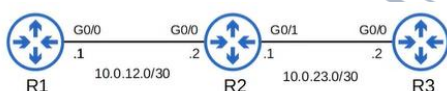
- Devices always prefer lower stratum servers (closer to source)

Configuring a Router as NTP Master

If a device is already syncing to an NTP server, meaning it is an NTP client it automatically acts as an NTP server too and other devices can sync to it. But what if there is no NTP server to sync to? You probably still want the devices in the network to have the same time, even if it is slightly inaccurate compared to the actual time.

So, how can you manually configure a Cisco device to operate as an NTP server, even though it isn't synced to another NTP server? This is the Command

Topology:



```
R1(config)#ntp ?
access-group      Control NTP access
allow             Allow processing of packets
authenticate      Authenticate time sources
authentication-key Authentication key for trusted time sources
broadcastdelay    Estimated round-trip delay
clock-period      Length of hardware clock tick
logging           Enable NTP message logging
master            Act as NTP master clock
max-associations  Set maximum number of associations
maxdistance       Maximum Distance for synchronization
mindistance       Minimum distance to consider for clockhop
orphan            Threshold Stratum for orphan mode
panic             Reject time updates > panic threshold (default 1000Sec)
passive           NTP passive mode
peer             Configure NTP peer
server            Configure NTP server
source            Configure interface for source address
trusted-key       Key numbers for trusted time sources
update-calendar   Periodically update calendar with NTP time
```

If no external NTP server is available, a router can act as a **local time source**.

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```
R1(config)#ntp master ?
<1-15> Stratum number
<cr>

R1(config)#ntp master
R1(config)#do show ntp associations

  address      ref clock      st  when  poll reach  delay  offset  disp
*~127.127.1.1  .LOCL.         7   2    16   377  0.000  0.000  0.292
* sys.peer, # selected, + candidate, - outlyer, x falseticker, ~ configured
R1(config)#do show ntp status
Clock is synchronized, stratum 8, reference is 127.127.1.1
...
```

Explanation

- ntp master makes the router act as an NTP server
- It uses itself as the reference clock (127.127.1.1)

Important details:

- 127.127.1.1 is a loopback address (internal only)
- Default stratum of ntp master = 8
- Association table shows 7, but actual device stratum = 8

This allows all devices in the network to share the same time, even if it's not perfectly accurate.

Remember, the entire 127.0.0.0/8 address range is reserved for loopback addresses.

Loopback addresses and loopback interfaces are different concepts, although the names are similar, so don't confuse the terms, Loopback interfaces are virtual interfaces in the router and their addresses can be advertised to other devices using OSPF etc.

Loopback addresses are a totally different concept, these addresses are totally internal to the local device and can't be reached by other devices.

Clients Syncing to NTP Master

After configuring R1 as NTP master, other routers use it as their time source.

```
R2(config)#ntp server 10.0.12.1
R2(config)#do show ntp associations

  address      ref clock      st  when  poll reach  delay  offset  disp
*~10.0.12.1    127.127.1.1      8   2    64    1  5.263  62.494  187.64
* sys.peer, # selected, + candidate, - outlyer, x falseticker, ~ configured

R3(config)#ntp server 10.0.12.1
R3(config)#do show ntp associations

  address      ref clock      st  when  poll reach  delay  offset  disp
*~10.0.12.1    127.127.1.1      8  45    64   17 21.534 -21.440  0.976
* sys.peer, # selected, + candidate, - outlyer, x falseticker, ~ configured
```

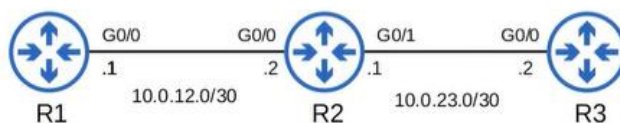
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Explanation:

- Both R2 and R3 synchronize with R1
- R1 acts as a central time source
- Even without Internet access, all devices maintain consistent time

NTP Peering (Symmetric Active Mode)

Devices at the same stratum can peer with each other.



```
R2(config)#ntp peer 10.0.23.2
R2(config)#do show ntp associations
address      ref clock      st   when   poll reach  delay  offset  disp
*~10.0.12.1   127.127.1.1     8    60     64    17 24.040 206.682 0.987
~10.0.23.2    10.0.12.1       9    33     64    0  0.000  0.000 15937.
* sys.peer, # selected, + candidate, - outlyer, x falseticker, ~ configured
```

```
R3(config)#ntp peer 10.0.23.1
R3(config)#do show ntp associations
address      ref clock      st   when   poll reach  delay  offset  disp
*~10.0.12.1   127.127.1.1     8    11     64    37 12.605 -7.406 63.575
~10.0.23.1    10.0.12.1       9    1      64    0  0.000  0.000 15937.
* sys.peer, # selected, + candidate, - outlyer, x falseticker, ~ configured
```

Explanation:

- R2 and R3 are at same stratum (9)
- They form a peer relationship

Benefits:

- Improve time accuracy
- Provide backup if R1 becomes unreachable

Both still use R1 as their primary reference but can rely on each other if needed.

Computer Networks - CCNA

NTP Authentication

NTP authentication ensures that devices only synchronize with trusted servers.

NTP authentication can be configured, although it is optional.

It allows NTP clients to ensure they only sync to the intended servers.

To configure NTP authentication:

`ntp authenticate`

`ntp authentication-key key-number md5 key`

`ntp trusted-key key-number`

`ntp server ip-address key key-number`

Enable NTP authentication

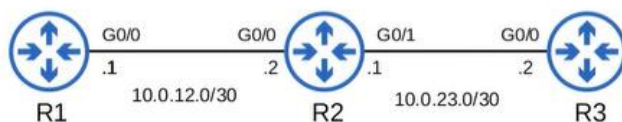
Create the NTP authentication key(s)

Specify the trusted key(s)

Specify which key to use for the server

This command isn't needed on the server (R1).

Now let's configure these on R1, R2, R3



CLI Output

```
R1(config)#ntp authenticate
```

```
R1(config)#ntp authentication-key 1 md5 jeremysitlab
```

```
R1(config)#ntp trusted-key 1
```

```
R2(config)#ntp authenticate
```

```
R2(config)#ntp authentication-key 1 md5 jeremysitlab
```

```
R2(config)#ntp trusted-key 1
```

```
R2(config)#ntp server 10.0.12.1 key 1
```

```
R2(config)#ntp peer 10.0.23.2 key 1
```

```
R3(config)#ntp authenticate
```

```
R3(config)#ntp authentication-key 1 md5 jeremysitlab
```

```
R3(config)#ntp trusted-key 1
```

```
R3(config)#ntp server 10.0.12.1 key 1
```

```
R3(config)#ntp peer 10.0.23.1 key 1
```

Computer Networks - CCNA

Explanation

- Authentication is optional but increases security
- Devices only sync if:
 - They share the same key number
 - They use the same password

Steps involved:

1. Enable authentication
2. Create a key (number + password)
3. Mark key as trusted
4. Apply key to NTP server/peer

Note: The ntp server ... key command is not required on the server itself

